

# Circles and Quadrilaterals

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## 1 Introduction

Squares, rectangles, trapezoids — all of these are the "special" quadrilaterals that we learn as elementary or middle school students. Most of them have distinct properties but there are sets of quadrilaterals that are considered more "specific" or "general" because they possess the properties of other quadrilaterals. Although this may seem obvious, this is also true for a more expansive range of quadrilaterals than the simple groups. This article contains some such quadrilaterals that have relations to circles, such as *cyclic quadrilaterals*, *tangential quadrilaterals*, *orthodiagonal quadrilaterals*, and the most specific of these, which contains a mixture of properties from each, *bicentric quadrilaterals*.

## 2 Cyclic Quadrilaterals

### 2.1 Definition

**Definition** (Cyclic Quadrilateral). A convex quadrilateral circumscribed within a circle.

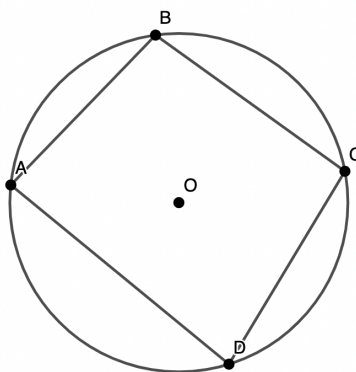


Figure 1: Cyclic quadrilateral

## 2.2 Properties

### 2.2.1 Supplementary Angles

In a cyclic quadrilateral, two opposite angles are supplementary.

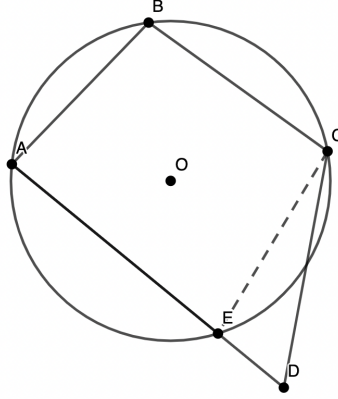


Figure 2: Supplementary angles

Consider quadrilateral  $ABCD$ . What we want to show is that the two opposite angles sum to  $180^\circ$ . First, let the center of the circle be  $O$ . Let  $\angle ABC$  be  $\alpha$  and  $\angle ADC$  be  $\beta$ . Constructing  $\angle AOC$ , we see that  $\angle AOC = 2\alpha$  because the angle subtended by an arc of a circle at its center is twice the angle it subtends anywhere on the circle's circumference. Following the same reasoning,

$$\begin{aligned} 360^\circ - \angle AOC &= 2\beta \\ \angle AOC + 360^\circ - \angle AOC &= 360^\circ \\ 2\alpha + 2\beta &= 360^\circ \\ \alpha + \beta &= 180^\circ \\ \angle ABC + \angle ADC &= 180^\circ \end{aligned}$$

The other two angles must be supplementary for the same reasoning.

### 2.2.2 Ptolemy's Theorem

In a cyclic quadrilateral, the product of its sides is equal to the sum of the product of its two pairs of opposite sides. With cyclic quadrilateral  $ABCD$ , this means  $AC \cdot BD = AB \cdot CD + AD \cdot BC$ .

Consider cyclic quadrilateral  $ABCD$ .  $AB$  is opposite to  $CD$  and  $AD$  is opposite to  $BC$ . Construct diagonals  $AC$  and  $BD$ . Because they subtend the same arcs,  $\angle BAC = \angle BDC$  and  $\angle ADB = \angle ACB$ . Construct  $E$  on  $AC$  such that  $\angle ABE = \angle CBD$ . It follows that  $\triangle ABE \sim \triangle DBC$  and  $\triangle EBC \sim \triangle ABD$ . Therefore,  $\frac{AE}{AB} = \frac{DC}{DB}$  and  $\frac{EC}{BC} = \frac{AD}{BD}$ . This gives

$$AE \cdot DB = AB \cdot DC$$

and

$$EC \cdot BD = BC \cdot AD,$$

and because  $AE + EC = AC$ , we get  $AC \cdot BD = AB \cdot CD + AD \cdot BC$ .

### 2.2.3 Brahmagupta's Formula

The area  $S$  of a cyclic quadrilateral with sides  $a, b, c, d$  is given by  $S = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ , where  $s$  is the semi-perimeter of the quadrilateral.

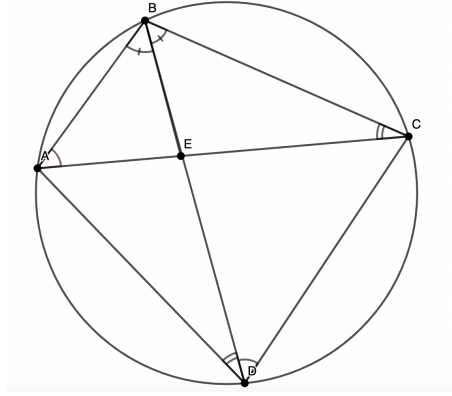


Figure 3: Ptolemy's Theorem

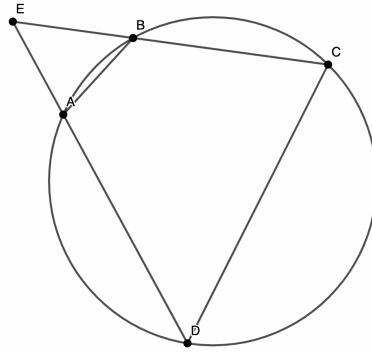


Figure 4: Brahmagupta's Formula

One can see that this is very similar to Heron's formula for a triangle. This observation will be used in the proof.

Consider quadrilateral  $ABCD$ , with  $AB = a$ ,  $BC = b$ ,  $CD = c$ , and  $DA = d$ . Extend the non-parallel opposite lines to a point  $E$  outside of the circumcircle. Denote  $CE = e$  and  $DE = f$ . This proof assumes that these two sides are  $AD$  and  $BC$ . Heron's formula can now be used on  $\triangle CDE$ .

$$[CDE] = \sqrt{\frac{1}{16}(c+e+f)(-c+e+f)(c+e-f)(c-e+f)},$$

so

$$4[CDE] = \sqrt{(c+e+f)(-c+e+f)(c+e-f)(c-e+f)}.$$

We also have that  $\angle BCD$  is supplementary to  $\angle BAD$ , which is supplementary to  $\angle BAE$ , meaning  $\angle BCD \cong \angle BAE$ . For similar reasoning,  $\angle CDA \cong \angle ABE$ . Therefore,  $\triangle ABE \sim \triangle CDE$ , meaning

$$\frac{[ABE]}{[CDE]} = \frac{a^2}{b^2},$$

and since  $[ABE] = S = c^2 - a^2$ ,

$$\frac{S}{[CDE]} = \frac{c^2 - a^2}{a^2}.$$

Additionally,  $\triangle ABE \sim \triangle CDE$ , meaning that  $\frac{e}{c} = \frac{f-d}{a}$  and  $\frac{f}{c} = \frac{e-b}{a}$ , which gives  $e+f = c \cdot \frac{b+d}{c-a}$  and  $e-f = c \cdot \frac{b-d}{c+a}$  also due to similar triangles. Substituting these into the original Heron's formula results in

the four expressions:

$$2c \cdot \frac{s-a}{c-a}, 2c \cdot \frac{s-c}{c-a}, 2c \cdot \frac{s-d}{c+a}, 2c \cdot \frac{s-b}{c+a}.$$

Therefore,  $[CDE] = \frac{c^2}{c^2-a^2}(\sqrt{(s-a)(s-b)(s-c)(s-d)})$ , and

$$\begin{aligned} S &= [CDE] \cdot \frac{c^2-a^2}{a^2} \\ &= \frac{c^2-a^2}{a^2} \cdot \frac{c^2}{c^2-a^2} \cdot \sqrt{(s-a)(s-b)(s-c)(s-d)} \\ &= \sqrt{(s-a)(s-b)(s-c)(s-d)} \end{aligned}$$

### 3 Tangential Quadrilaterals

#### 3.1 Definition

**Definition** (Tangential Quadrilateral). A convex quadrilateral that contains an incircle that is tangent to all 4 sides of the quadrilateral.

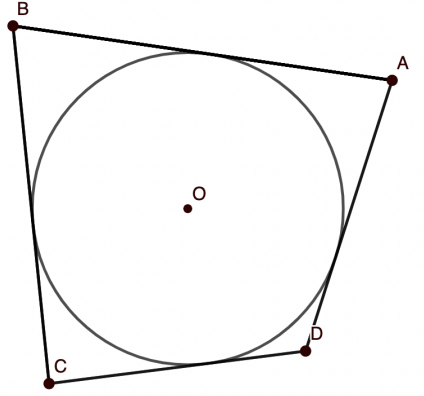


Figure 5: Tangential quadrilateral

#### 3.2 Properties

##### 3.2.1 Pitot's Theorem

The two sums of lengths of opposite sides are equal. In tangential quadrilateral  $ABCD$ ,  $AB + CD = AD + BC$ .

The proof for this theorem is very quick. Since the sides are composed of tangent lines, let the points of tangency on segment  $AB$  be  $P$ ,  $Q$  on  $BC$ ,  $R$  on  $CD$ , and  $S$  on  $AD$ . It follows that  $AP = AS$ ,  $BP = BQ$ ,  $DS = DR$ ,  $CR = CQ$ . Labeling  $AP$  as  $p$ ,  $BP$  as  $q$ ,  $DQ$  as  $r$ , and  $CR$  as  $s$ , From there, it can be seen that

$$\begin{aligned} AB &= AP + BP = p + q \\ CD &= CR + DR = r + s \\ BC &= BQ + CQ = q + s \\ AD &= AS + DS = p + r. \end{aligned}$$

Therefore,  $p + q + r + s = q + s + p + r$ , which implies that

$$AP + BP + CR + DR = BQ + CQ + AS + DS$$

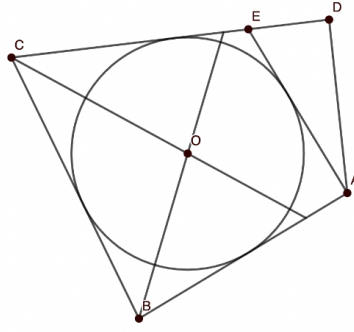


Figure 6: Pitot's Theorem

so  $AB + CD = AD + BC$ .

There is also a converse to this theorem: if there exists a convex quadrilateral  $ABCD$  with  $AB + CD = AD + BC$ , then a semicircle tangent to all 4 sides can be inscribed within it.

Consider a convex quadrilateral  $ABCD$ . Instead of making a circle tangent to all 4 sides, which would be counterproductive to the purpose of the proof, construct a circle tangent to  $AB$ ,  $BC$ , and  $CD$  using angle bisectors of  $\angle B$  and  $\angle C$ . This can be done for the same reasoning as why an incircle can be constructed using angle bisectors. The angle bisectors will split the shape into triangles with equal length altitudes, which are the points of tangency.

The most standard proof for this is proof by contradiction.

If we assume that the circle is not tangent to  $AD$ , we can draw a tangent line through  $A$  to a point  $E$  lying on segment  $CD$ . By the previously proven Pitot's theorem, we have  $AB + CE = AE + BC$ . However, we also have  $AB + CD = AD + BC$ . Subtracting the two:

$$CD - CE = AD - AE.$$

The second is subtracted from the first to ensure that negative side lengths don't occur. We also have that

$$CD - CE = ED = AD - AE.$$

These are all part of  $\triangle AED$ , and because it is not true that the sides are greater than or equal to the sum of any other two, the triangle must be degenerate. Therefore, the original assumption was false, and it must be true that  $AD$  is tangent to  $ABCD$ . The circle is tangent to all 4 sides of  $ABCD$ .

It can also easily be seen from this that  $AB + CD$  and  $AD + BC$  are both equal to the semi-perimeter.  $p + q + r + s = S$ , the semiperimeter.

### 3.2.2 Area

Area can be expressed as  $A = rs$ , where  $r$  is the inradius and  $s$  is the semiperimeter. The reasoning behind this claim is nearly identical to that of a triangle with an incircle, especially since the areas can be expressed in the same way (but with a different semiperimeter value).

Consider  $O$ , the center of the incircle of quadrilateral  $ABCD$ . Segments can be drawn to the nearest vertices. This will result in 4 triangles:  $\triangle AOD$ ,  $\triangle BOA$ ,  $\triangle COB$ , and  $\triangle DOC$ . All of these have an altitude of  $r$ , meaning that their total areas combined can be expressed as  $\frac{1}{2}ra + \frac{1}{2}rb + \frac{1}{2}rc + \frac{1}{2}rd$ , where  $a$ ,  $b$ ,  $c$ , and  $d$  are sides of the quadrilateral. This can be rewritten as  $\frac{1}{2}r(a + b + c + d)$ , or  $rs$ , where  $s$  is the semiperimeter. Because the triangles partitioned the quadrilateral, we get  $A = rs$ .

## 4 Orthodiagonal Quadrilaterals

### 4.1 Definition

**Definition** (Orthodiagonal Quadrilateral). A convex quadrilateral whose diagonals intersect at right angles.

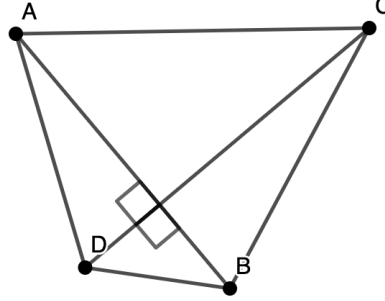


Figure 7: Orthodiagonal quadrilateral

### 4.2 Properties

#### 4.2.1 Sum of Squares of Diagonals

A convex quadrilateral  $ABCD$  is orthodiagonal iff  $AB^2 + CD^2 = BC^2 + DA^2$ .

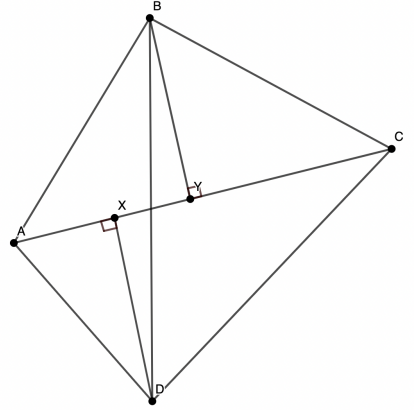


Figure 8: Property of orthodiagonal quadrilateral

Construct convex quadrilateral  $ABCD$  and its diagonals,  $AC$  and  $BD$ . Drop altitudes from points  $D$  and  $B$  to  $AC$ , calling them  $x$  and  $y$  respectively. The figure can then be partitioned into 4 right triangles with the Pythagorean properties:

$$AY^2 + BY^2 = AB^2$$

$$BY^2 + CY^2 = BC^2$$

$$CX^2 + DX^2 = CD^2$$

$$AX^2 + DX^2 = DA^2$$

We can notice that some of these equations share terms, such as  $BY^2$  and  $DX^2$ . Eliminating these from each other, we get

$$AB^2 - BC^2 + CD^2 - DA^2 = AY^2 - CY^2 + CX^2 - AX^2$$

The right hand side can be rearranged to:

$$\begin{aligned}
AY^2 - AX^2 + CX^2 - CY^2 &= (AY + AX)(AY - AX) + (CX + CY)(CX - CY) \\
&= (AY + AX)XY + (CX + CY)XY \\
&= (AX + CX + AY + CY)XY \\
&= 2(AC)XY.
\end{aligned}$$

We want the distance between  $x$  and  $y$  to be 0 so that the perpendiculars intersect at the intersection of the diagonals. Setting  $XY$  to 0:

$$AB^2 - BC^2 + CD^2 - DA^2 = 0,$$

so  $AB^2 + CD^2 = BC^2 + DA^2$ .

#### 4.2.2 Sum of Angles of Diagonals

Another property is a convex quadrilateral  $ABCD$  is orthodiagonal iff  $\angle OAB + \angle OBA + \angle OCD + \angle ODC = 180^\circ$ , where  $O$  is the intersection of the diagonals.

The sum  $\angle OAB + \angle OBA + \angle OCD + \angle ODC = 2\pi - 2\theta$ , where  $\theta$  is the angle between the diagonals (hence, the angles excluded from the theorem). Because we want theta to be equal to  $90^\circ$ ,  $2\theta = 180^\circ$ , and therefore,  $\angle OAB + \angle OBA + \angle OCD + \angle ODC = 180^\circ$ .

There are a lot of properties to orthodiagonal quadrilaterals, but most of them revolve around similar reasoning to theorem 1 and are of the same form. The circumradii, medians of triangles within the quadrilateral, and reciprocals of the altitudes are all related in the same proportion.

## 5 Bicentric Quadrilaterals

### 5.1 Definition

**Definition** (Bicentric Quadrilateral). A convex quadrilateral that contains both a circumcircle and an incircle; satisfies both properties of cyclic and tangential quadrilaterals.

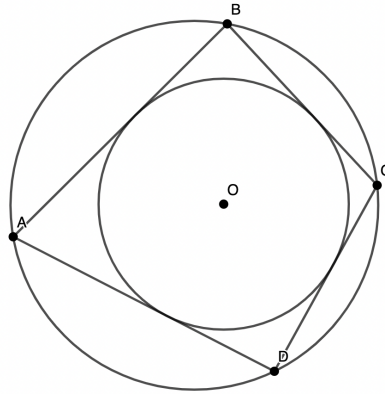


Figure 9: Bicentric quadrilateral

## 5.2 Properties

### 5.2.1 Area

A bicentric quadrilateral has area  $\sqrt{abcd}$ , where  $a$ ,  $b$ ,  $c$ , and  $d$  are its sides.

Consider a bicentric quadrilateral. Since, by definition, it is a quadrilateral that contains properties of both cyclic and tangential quadrilaterals, Brahmagupta's and Pitot's theorems can simply be algebraically combined to find the area of a bicentric quadrilateral.  $S = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ , so

$$16S^2 = (-a+b+c+d)(a-b+c+d)(a+b-c+d)(a+b+c-d).$$

We have that  $a+c=b+d$ , so

$$a = -c + b + d$$

$$b = a + c - d$$

$$c = -a + b + d$$

$$d = a + c - b$$

Plugging in  $a$  for  $-a$ ,  $b$  for  $-b$  and so on,

$$16S^2 = (2a)(2b)(2c)(2d),$$

so  $S = \sqrt{abcd}$ .

## 6 Special quadrilaterals

Some of the standard groups of shapes are special cases of the ones listed above. *Squares*, because they have all of the properties that satisfy cyclic, tangential, and even orthodiagonal quadrilaterals, are bicentric. It can easily be seen that the area formula indeed does satisfy a square, as  $A = s^2$  where  $s$  = side and  $A = \sqrt{s^4}$  are equivalent.

Given that all of the following kinds of quadrilaterals satisfy the property that their angles are supplementary, rectangles, isosceles trapezoids, and kites with two right angles are cyclic.

Although rhombuses aren't cyclic, they are tangential because of their paired sides. Kites are also tangential, orthodiagonal, and cyclic, making them bicentric.

## 7 Resources

"Brahmagupta's Formula and Theorem."

<https://www.cut-the-knot.org/Generalization/Brahmagupta.shtml>.

"Cyclic Quadrilateral." Byju's.

<https://byjus.com/maths/cyclic-quadrilateral/>.

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